Sahaj Bhadja

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Summary

Associate software engineer with 2+ years building **operator-style interfaces**, backend services, tests, CI workflows, and Linux-friendly tooling; eager to ship field-tested software for autonomous battery-electric rail systems.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built **Python-backed platform workflows** for 500+ users, translating requirements into APIs, validation logic, monitored jobs, technical documentation, release notes, and maintainable delivery through reviewed changes and support runbooks.
- Improved **service reliability and debugging** by tuning indexed queries, background jobs, and CI/CD releases, reducing dashboard latency from **1.2s to 350ms** with traceable production handoffs and release notes.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed **Python and JavaScript services** handling 10,000+ daily requests, implementing REST APIs, file ingestion, responsive interfaces, automated checks, documented fixes, and sprint-ready production workflows with code review.
- Supported **root-cause testing workflows** by reproducing OCR, persistence, and execution defects, documenting regression cases, reviewing fixes, and cutting instructor feedback turnaround by **60%** across validation cycles and QA notes.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Created **Python evaluation pipelines** across 100,000+ records, producing reproducible metrics, logs, reports, and review artifacts for debugging model behavior and validating decision workflows under experiment constraints.
- Documented **algorithm behavior and edge cases** for researchers, turning ad hoc experiments into repeatable test scenarios with traceable outputs, clearer handoffs, and reviewable conclusions for integration planning.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered **cross-platform services and ETL workflows** using Python, SQL, AWS Glue, and S3, improving response time by **30%** across integrations, releases, reviews, and handoffs for distributed users.
- Maintained **20+ automated test suites** using Jenkins, PyTest, and Mockito across validation flows, increasing CI reliability by **68%** and reducing release regressions during coordinated deployments and reviews.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built **operator-facing application modules** with REST APIs, WebSockets, and SQL/NoSQL persistence, supporting 2M+ synchronized records across dashboards, real-time workflows, integrations, reviews, releases, support needs, and field logs.
- Automated **Docker-based build and test flows** with Jenkins, cutting release time by **50%** while improving deployment consistency, troubleshooting documentation, and shared-environment handoffs across staging workflows and release checks.

Skills

Languages: Python, JavaScript, TypeScript, Java, C++, SQL, Bash, C# familiarity, Dart familiarity
UI/HMI: React, Next.js, Flutter familiarity, Android concepts, responsive interfaces, dashboards, operator workflows
Backend: REST APIs, Flask, FastAPI, Node.js, service design, data ingestion, root-cause debugging, integration testing
Systems: Linux CLI, Docker, Git branching, pull requests, code review, CI/CD, GitHub Actions, Jenkins
Data/AI: PostgreSQL, MySQL, MongoDB, SQL Server, ETL, validation checks, AI-assisted coding, test automation

Projects

CPU Scheduling Simulator Live | *C++, React.js, JavaScript*
- Implemented **FCFS, SJF, SRT, RR, priority, and MLQ scheduling** with metric calculation and visualization, strengthening systems fundamentals, debugging discipline, UI-to-algorithm correctness, real-time reasoning, and automated test coverage.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*
- Built an **AI-powered SQL workspace** supporting 10+ databases, accelerating query authoring, schema exploration, and database management through Java services and TypeScript interfaces with Docker workflows and Git-based collaboration.

Linguistix Code | *Python, AWS, Docker, CI/CD*
- Developed a **containerized REST processing API** with AWS deployment, CI/CD releases, validation checks, and reproducible execution, achieving **87%** classification accuracy with diagnostic reporting, automated tests, and logs.